AI Student Dropout Risk Prediction System – Hackathon Guide

1. Understanding the Problem

The goal is to build an AI system that predicts which students are at risk of dropping out. The system should analyze academic, behavioral, and engagement data to identify students who may need early intervention from teachers or institutions.

2. Data Required

The model can use the following data: • Academic Data: Grades, GPA, assignment scores, exam performance • Behavioral Data: Attendance percentage, class participation • Personal Factors: Financial background, distance from college, family support • Engagement Data: LMS activity, library usage, portal logins

3. AI / Machine Learning Model

Use machine learning models such as: • Logistic Regression • Decision Trees • Random Forest • XGBoost Python libraries that can be used: Pandas, NumPy, Scikit■learn, TensorFlow or PyTorch.

4. System Features

Create a simple web dashboard that allows: • Uploading student data (CSV file) • Predicting dropout risk using AI • Displaying students by risk level • Showing recommendations for teachers or administrators

5. Innovation Ideas

To stand out in the hackathon, add extra features such as: • Early warning alerts for high■risk students • AI recommendations for interventions • Chatbot for student support • Visualization dashboard for institutions • Personalized learning support suggestions

6. Demo Flow

During the hackathon presentation: 1. Upload student dataset 2. Run the AI prediction model 3. Display the dashboard with risk levels 4. Show suggested interventions such as mentoring or counseling

7. Judging Focus

Judges typically evaluate: • Innovation • Real■world impact • Working prototype • Data usage and AI explanation • Clear UI/dashboard

Example Output

Student	Risk Score	Risk Level
Student A	0.23	Low
Student B	0.65	Medium
Student C	0.89	High